Fiscal Year:	FY 2025	**Task Last Updated:**	FY 01/27/2025
PI Name:	Allaway, Heather Ph.D.		
Project Title:	A Time Course of Bone Microarchitectural and Material Property Changes in Male and Female Mice During Simulated Unloading and Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	hallaway@lsu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	321-431-8318
Organization Name:	Louisiana State University and A&M College		
PI Address 1:	Kinesiology		
PI Address 2:	2139 Huey P Long Field House		
PI Web Page:			
City:	Baton Rouge	**State:**	LA
Zip Code:	70803-0001	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2023 HERO NNJ23ZSA001N-OMNIBUS : NASA Human Research Program Omnibus Opportunity
Start Date:	03/01/2024	**End Date:**	02/28/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 02/28/2026 per S. Mack-Phillips/NASA JSC (Ed., 2/12/2026).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lau, Anthony Ph.D. (The College of New Jersey)		
Grant/Contract No.:	80NSSC24K0406		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The evaluation of sex differences in the time course of physiologic responses and adaptations to the space environment is critical to ensuring the safety of crews as the world prepares for the first crewed spaceflights beyond low Earth orbit in 50 years and the first woman to step on the Moon. Microgravity during spaceflight damages musculoskeletal tissues, but the understanding of sex differences in musculoskeletal atrophy during spaceflight is incomplete. Maintaining musculoskeletal health during long-duration spaceflight is crucial for ensuring astronauts are physically ready for exploration class missions and mission success upon return to partial weightbearing activity. Due to the known links between sex and musculoskeletal health, it is critical to get a better understanding of “how” and “why” bone microarchitectural and material property changes occur over a mission duration and if that change occurs differently in males and females. Finite element (FE) modeling protocols and computational models have been developed to better identify how bone strength is compromised during reduced weight-bearing and other challenges. Preliminary data using FE modeling on female mice flow on Space Shuttle mission STS-135 indicated that proximal tibia bone density loss (-17%) underpredicted the attenuation in structural strength (-34%). These findings demonstrate a limitation in the current clinical assessments of bone health and fracture risk using dual energy x-ray absorptiometry (DXA), as well as the standard micro computed tomography (microCT) analysis for predicting bone health during spaceflight. The objective of the current proposal is to capitalize on samples from two completed projects that the team has already acquired through tissue sharing opportunities. The parent projects had male and/or female mice that underwent unloading (hindlimb unloading or spaceflight) for 14, 28, or 38 days (and appropriate controls). We will quantify bone cellular, microarchitecture, and strength changes using histology, microCT, computational FE modeling, and material property characterization. Successful completion of this project will provide foundational data for further mechanistic and translational studies examining sex differences in the musculoskeletal adaptation to spaceflight environmental stressors that will enable more specific countermeasure development.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The work is significant to NASA’s goal for maximizing astronaut health during long-duration spaceflight. The proposed work will provide critical insight on how changes in bone microarchitecture and material properties during spaceflight contribute to overall bone strength, which are important factors for understanding and mitigating fracture risk. The proposed work will synergize with the development of novel imaging modalities to monitor bone health and countermeasure development, which will benefit astronauts and people on Earth.
Task Progress:	The proposed work uses a multi-scale approach to assess changes in bone health and strength in already available hindlimb bone specimens obtained from tissue-sharing opportunities: 1) male mice from a 38-day International Space Station (ISS) flight study (80NSSC19K0392) and 2) male and female mice from a HU time course (14 vs 28 days) study (80NSSC19K1025). All live animal work was completed by July 2022 under approved Institutional Animal Care and Use Committee (IACUC) protocols at the parent study institutions. In the past reporting period, we have made progress to completion of SA1, but have been delayed due to equipment issues and study staff illness which limited training time. SA1: Determine the time course of bone microarchitectural and material property changes from long-duration unloading and the subsequent impact on bone strength. Calcified and decalcified bone histology: All samples that are to be embedded in plastic resin have been embedded. A few samples will need to be re-embedded due to extreme shifts in bone angle during the solidification time of embedding. This has been delayed due to illness in the graduate student trainee. 2/3 of the samples have been decalcified and are paraffin-embedded. The last set of samples is currently undergoing decalcification. Graduate student training on microtoming, staining, and histomorphological quantification using the BioQuant software is underway. Student training on microtoming is going slow as we are being extremely cautious to not lose any precious mouse tissue samples due to a lack of skill confidence. Training on the Bioquant system was delayed due to computer issues that we are working with the BioQuant team to resolve. Ex vivo microCT Imaging & Analysis: uCT scanning was delayed due to equipment shutdown for May-August. This is a core facility and was an unanticipated shutdown for the summer. In the fall, 2/3 of the tibia scans have been completed. We anticipate starting scans of the femur samples in February. Assessment of the completed scans is underway to determine if any samples need to be rescanned. Finite Element (FE) Modeling: Nothing to report. Cannot begin FEA analysis until uCT scans are complete. Biomechanical Testing: Femoral cortical bone mechanical and material properties will be assessed using 3-point-bend-to-failure and spherical micro-indentation tests. Nothing to report. Cannot begin biomechanical testing until uCT scans are complete. When the uCT scans for a bone are complete the sample will be shipped for biomechanical testing. SA2: Evaluate sex differences in the time course of bone microarchitectural, material property, and strength changes from long-duration unloading. During the past reporting period, all researchers have remained blinded to the sex of the animals. Data collection has not progressed far enough to unblind the research team and evaluate this question.
Bibliography Type:	Description: (Last Updated: 01/28/2026)
Abstracts for Journals and Proceedings	Beard C, Lau AG, Bowman LC, Allaway HCM. "Methodological approaches to assess sex differences in the effects of hindlimb unloading on bone strength in mice." 2025 NASA Human Research Program Investigator's Workshop, Galveston Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigator's Workshop, Galveston Texas, January 28-31, 2025. , Jan-2025